E-HEALTH CARE MANAGEMENT SYSTEM

ABSTRACT

- The term electronic health record (EHR) refers to a patient's electronic data collection containing medical history that is accessible from any location by authorized personnel. E-healthcare has enhanced social and health outcomes and reduced medical errors. However, the security issues associated with the storage of patient data in IT frameworks pose an important obstacle to the growth of e-healthcare. The emergence of blockchain technology as a potential remedy to this problem has the potential to revolutionize the healthcare industry. Our work lays the foundation for secure patient data management using a blockchain-based approach, which incorporates a public ledger, private ledger, smart contracts, and context-based access control. The proposed architecture ensures reliable access to patient data, safe storage, and interoperability. Additionally, the proposed blockchain-based framework provides an efficient and trustworthy means of managing complex medical procedures. We also outlines potential uses for blockchain technology in the healthcare industry, such as facilitating secure and anonymous sharing of health data for research purposes.our proposed work a novel, accessible, interoperable, and audible approach for maintaining medical records using smart contracts.

INTRODUCTION

- As blockchain technology continues to gain traction, research in various sectors of science relevant to society is advanced thanks to the variety of applications it may be used for. Blockchain is seen as a revolutionary arrangement that resolves a few squeezing issues in the area of innovation, like the personality of the board, decentralization, information proprietorship, and trust-and information-driven decisionmaking. Each node in a blockchain-based distributed network is furnished with an indistinguishable duplicate of the information and is expected to comply with similar standards. Blockchain is a circulated record that grows as more information is contributed in the form of blocks. Seven ”V” traits define big data: volume, variety, velocity, variability, veracity, visualization, and value. The magnitude of the data may be described by its ”volume.” Data access speed is measured in velocity. Diverse sources and the unstructured character of the data are what we mean by variety. According to variability, the same data might have several interpretations. Data quality is a factor in veracity. Graphics, charts, and other plots are used in visualization to assist people to grasp the meaning of data and find out additional information.

OBJECTIVE

- The primary objective is to improve the security of electronic healthcare systems, particularly in the storage and management of Electronic Health Records (EHR), by leveraging the inherent security features of blockchain technology.
- Create a comprehensive and innovative blockchainbased framework that incorporates elements such as smart contracts, and contextbased access control. This framework aims to establish a secure, tamperresistant, and interoperable environment for managing patient data.
- Design the blockchain architecture to guarantee reliable access to patient data, allowing authorized personnel to retrieve information securely from any location at any time. The objective is to facilitate efficient and transparent data access while maintaining the integrity of the records.
- Focus on achieving interoperability within the healthcare system, enabling seamless communication and data exchange among different components. The project aims to enhance the overall efficiency of healthcare procedures and management through the integration of blockchain technology.

PROBLEM STATEMENT

- The proliferation of electronic health records (EHR) in e-healthcare has significantly improved health outcomes, but security concerns in storing patient data within IT systems hinder its growth. This study addresses this issue by proposing a blockchain-based solution, incorporating public and private ledgers, smart contracts, and context-based access control for secure, interoperable, and auditable medical record management.

LITERATURE SURVEY

- **MediBchain: A Blockchain Based Privacy Preserving Platform for Healthcare Data:**
- MediBchain: A Blockchain Based Privacy Preserving Platform for Healthcare Data | SpringerLink
- **ABSTRACT:** Healthcare data are grabbing the interest of cyber attackers in recent years. Annihilating consequences of healthcare data could be alleviated through decentralization. A peer to peer (P2P) network enables the property of decentralization, where different parties can store and run computation while keeping the sensitive health data private. Blockchain technology leverages decentralized or distributed process, which ensures the accountability and integrity of its use. This paper presents a patient centric healthcare data management system by using Blockchain as storage to attain privacy. Pseudonymity is ensured by using the cryptographic functions to protect patient's data.

LITERATURE SURVEY

- **Blockchain and Health IT: Algorithms, Privacy, and Data:**
- (PDF) Blockchain and Health IT: Algorithms, Privacy, and Data (researchgate.net)
- **ABSTRACT:** The President's Precision Medicine Initiative (PMI) is "enabling a new era of clinical care through research, technology, and policies that empower patients, researchers, and providers to work together toward the development of individualized care". 1 Its commitment to privacy and security in the setting of responsible data sharing and transparency is articulated in the "Privacy and Trust Principles" 1 and the "Data Security Policy Principles and Framework" 2 , developed by an interagency working groups including the Office of the National Coordinator for Health Information Technology in conjunction with multiple stakeholders. In this paper, we review the threats to the security, confidentiality, integrity, and availability of PMI data.

LITERATURE SURVEY

- **Blockchain for Secure EHRs Sharing of Mobile Cloud Based E-Health Systems:**
- Blockchain for Secure EHRs Sharing of Mobile Cloud Based E-Health Systems | IEEE Journals & Magazine | IEEE Xplore
- **ABSTRACT:** Recent years have witnessed a paradigm shift in the storage of Electronic Health Records (EHRs) on mobile cloud environments, where mobile devices are integrated with cloud computing to facilitate medical data exchanges among patients and healthcare providers. This advanced model enables healthcare services with low operational cost, high flexibility, and EHRs availability. However, this new paradigm also raises concerns about data privacy and network security for e-health systems. How to reliably share EHRs among mobile users while guaranteeing high-security levels in the mobile cloud is a challenging issue. In this paper, we propose a novel EHRs sharing framework that combines blockchain and the decentralized interplanetary file system (IPFS) on a mobile cloud platform. Particularly, we design a trustworthy access control mechanism using smart contracts to achieve secure EHRs sharing among different patients and medical providers.

LITERATURE SURVEY

- **Blockchain-based Personal Health Data Sharing System Using Cloud Storage:**

- Blockchain-based Personal Health Data Sharing System Using Cloud Storage | IEEE Conference Publication | IEEE Xplore

- **ABSTRACT:** With the advent of rapid development of wearable technology and mobile computing, huge amount of personal health-related data is being generated and accumulated on continuous basis at every moment. These personal datasets contain valuable information and they belong to and asset of the individual users, hence should be owned and controlled by themselves. Currently most of such datasets are stored and controlled by different service providers and this centralised data storage brings challenges of data security and hinders the data sharing. These personal health data are valuable resources for healthcare research and commercial projects. In this research work, we propose a conceptual design for sharing personal continuous-dynamic health data using blockchain technology supplemented by cloud storage to share the health-related information in a secure and transparent manner.

LITERATURE SURVEY

- **Security and privacy in electronic health records: A systematic literature review:**
- Security and privacy in electronic health records: A systematic literature review - ScienceDirect
- **ABSTRACT:** Objective To report the results of a systematic literature review concerning the security and privacy of electronic health record (EHR) systems. Data sources Original articles written in English found in MEDLINE, ACM Digital Library, Wiley InterScience, IEEE Digital Library, Science@Direct, MetaPress, ERIC, CINAHL and Trip Database. Study selection Only those articles dealing with the security and privacy of EHR systems. Data extraction The extraction of 775 articles using a predefined search string, the outcome of which was reviewed by three authors and checked by a fourth. Results A total of 49 articles were selected, of which 26 used standards or regulations related to the privacy and security of EHR data. The most widely used regulations are the Health Insurance Portability and Accountability Act (HIPAA) and the European Data Protection Directive 95/46/EC.

EXISTING SYSTEM

- In literature they introduced a novel blockchain-based data preservation system (DPS) for medical data. To provide a reliable storage solution to ensure the primitiveness and verifiability of stored data while preserving privacy for user. They leverage the blockchain framework. With their introduced DPS, users can preserve important data in perpetuity, and the originality of the data can be verified if tampering is suspected. In addition, they use prudent data storage strategies and a variety of cryptographic algorithms to guarantee user privacy; e.g., an adversary is unable to read the plain text even if the data are stolen. They implement a prototype of the DPS based on the real world blockchain-based platform Ethereum. Performance evaluation results demonstrate the effectiveness and efficiency of their system.

DISADVANTAGES OF EXISTING SYSTEM

- The existing work mentions preserving user privacy but doesn't provide detailed information about how privacy is maintained.
- The existing work does not highlight interoperability with other healthcare systems or platforms.
- While the existing work focuses on data preservation and tamper resistance, it may not efficiently handle the complex data management requirements in healthcare, such as patient consent management or data access control based on specific medical contexts.
- The existing work does not discuss the scalability of its system, which is essential in healthcare where large volumes of data are generated daily. Scalability is crucial for accommodating future growth and ensuring the system remains efficient.

PROPOSED SYSTEM

- We propose a block chain based smart health care system. We explores how blockchain technology may be integrated with healthcare records and data to facilitate secure and trustworthy data sharing. Our work lays the foundation for secure patient data management using a blockchain-based approach, which incorporates a public ledger, private ledger, smart contracts, and context-based access control. The proposed architecture ensures reliable access to patient data, safe storage, and interoperability. Additionally, the proposed blockchain-based framework provides an efficient and trustworthy means of managing complex medical procedures. We also outlines potential uses for blockchain technology in the healthcare industry, such as facilitating secure and anonymous sharing of health data for research purposes. Our proposed work a novel, accessible, interoperable, and audible approach for maintaining medical records using smart contracts.

ADVANTAGES OF PROPOSED SYSTEM

- Our work explicitly mentions incorporating features like context-based access control, which is likely to provide stronger privacy protection.
- Our work emphasizes interoperability, which is crucial for efficient data sharing in the healthcare industry.
- The proposed architecture in our work likely offers better management of complex healthcare data, including patient consent management and data access control based on specific medical contexts.
- Our work's emphasis on scalability ensures that the system can handle the increasing volume of healthcare data generated daily, making it a more sustainable solution for the future.

REQUIREMENTS

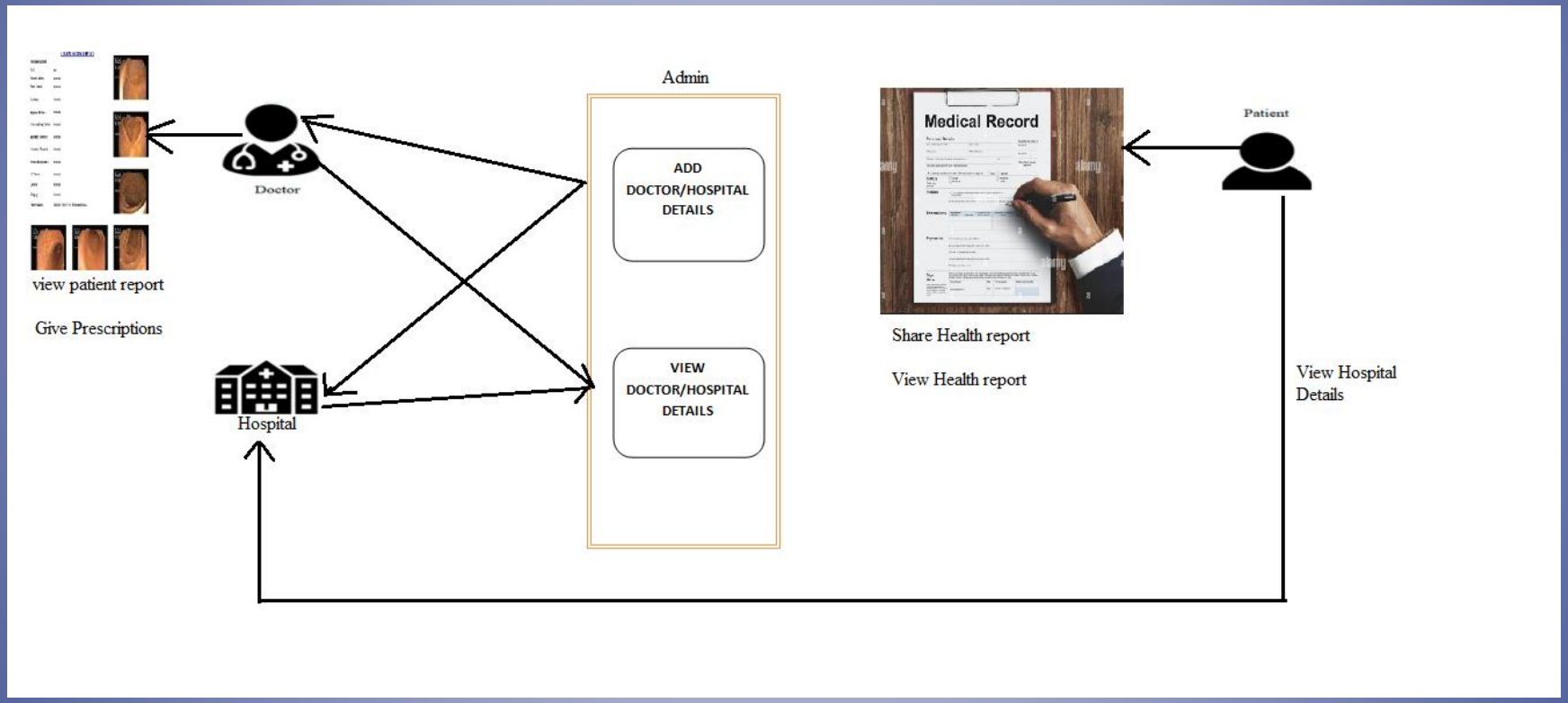
- **1) PYTHON IDLE (3.7.0)**
- **2) Node JS**
- **3) Visual Studio Community Version**
- **4) Ganache**
- **5) Metamask - Chrome extension**
- **6) LANGUAGES BACK-END: Python, Java Script, Solidity**
- **7) LANGUAGES FRONT-END: HTML, CSS, JS, Boot Strap**
- **8) FRAMEWORK: Flask**

SOFTWARE

1) Operating System : Windows Only
2) Processor : i5 and above
3) Ram : 8gb and above
4) Hard Disk : 25 GB in local drive

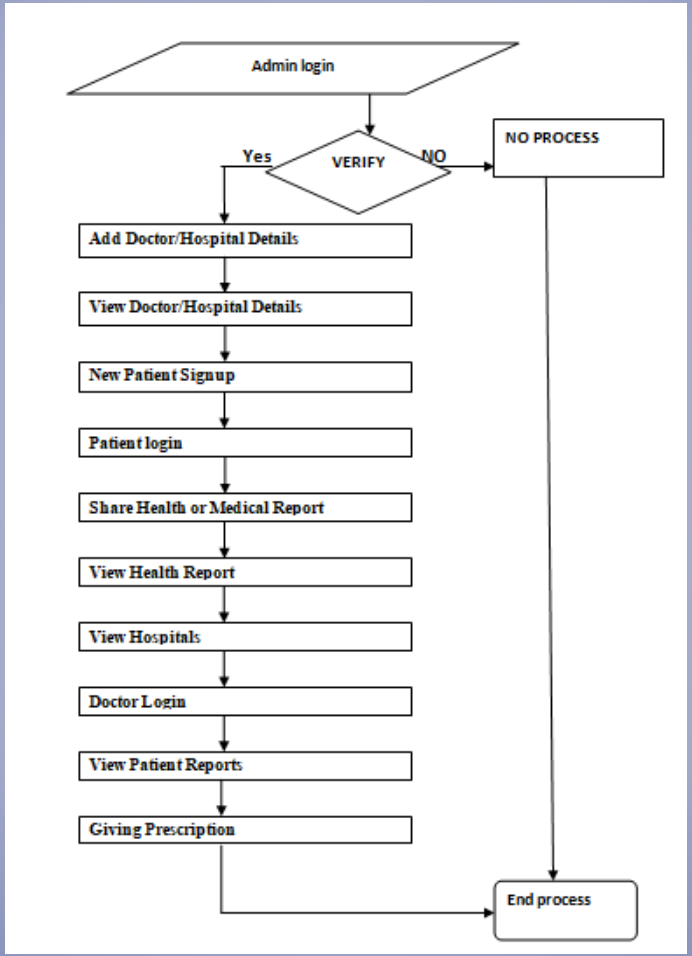
HARDWARE

SYSTEM ARCHITECTURE

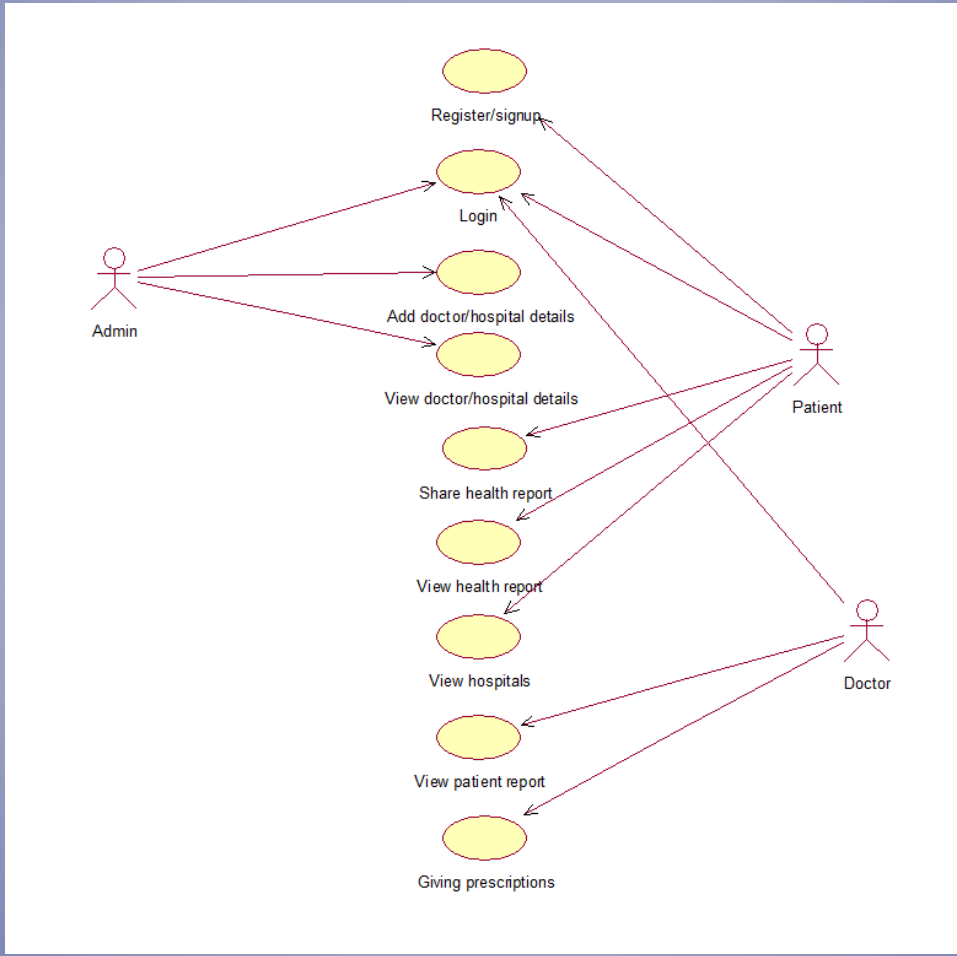


UML DIAGRAMS

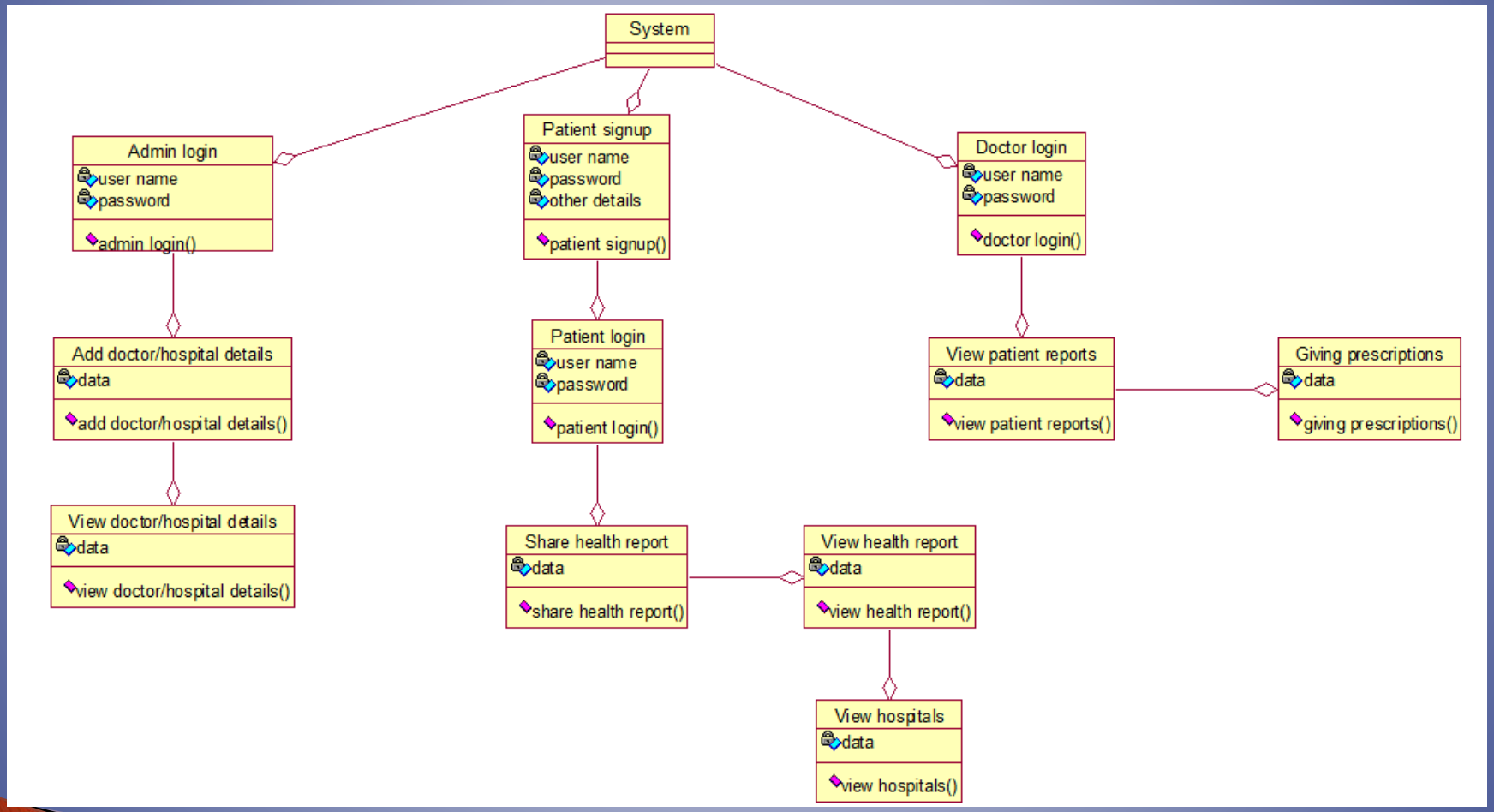
DATA FLOW DIAGRAM



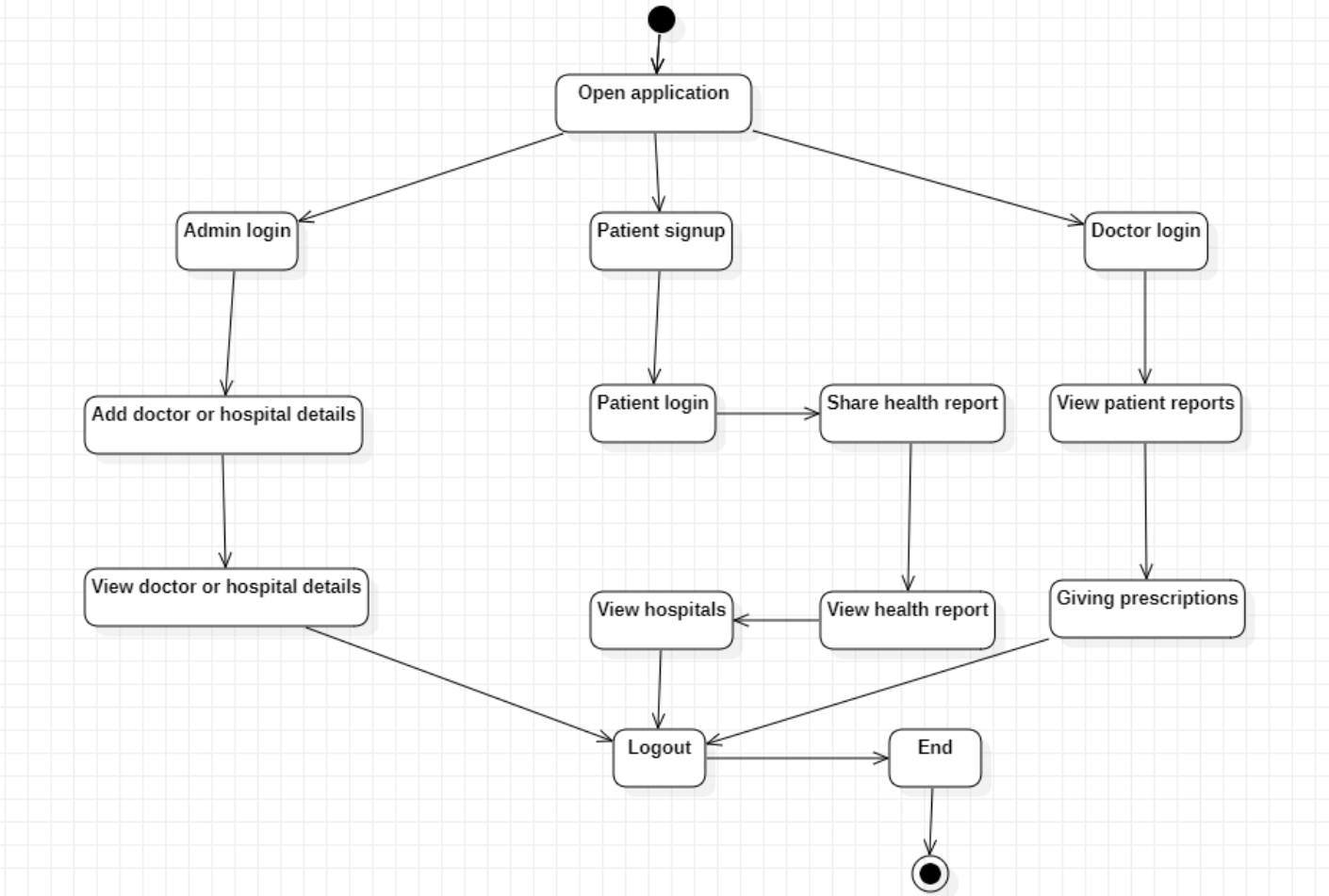
USE CASE DIAGRAM



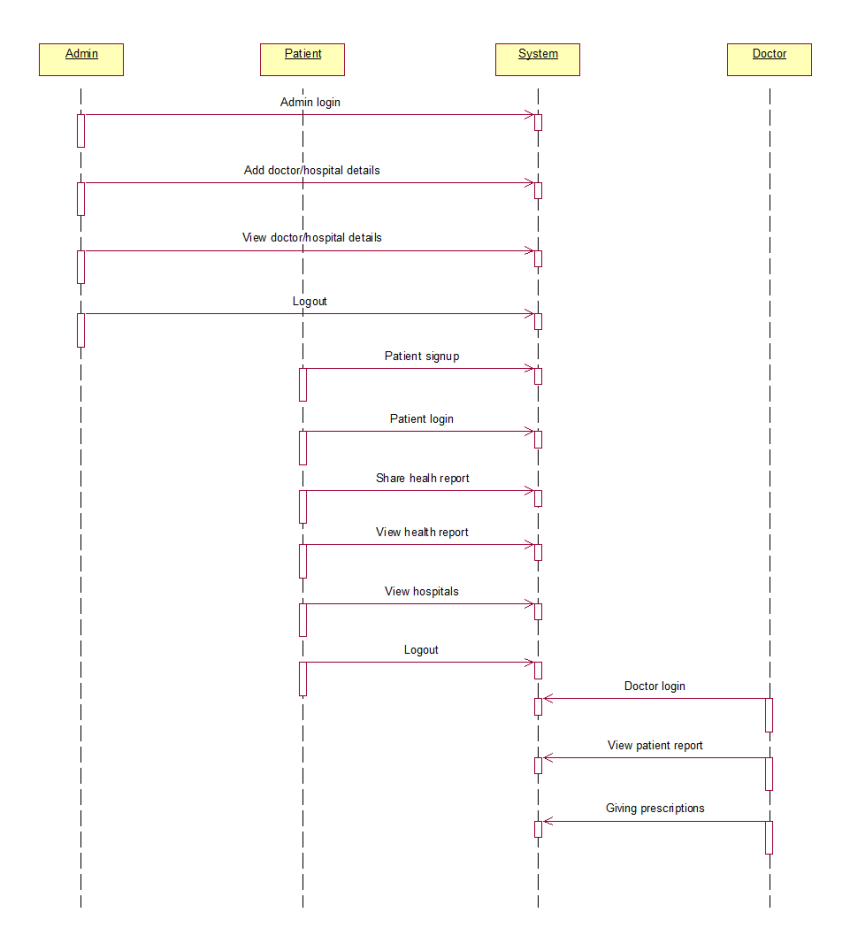
CLASS DIAGRAM



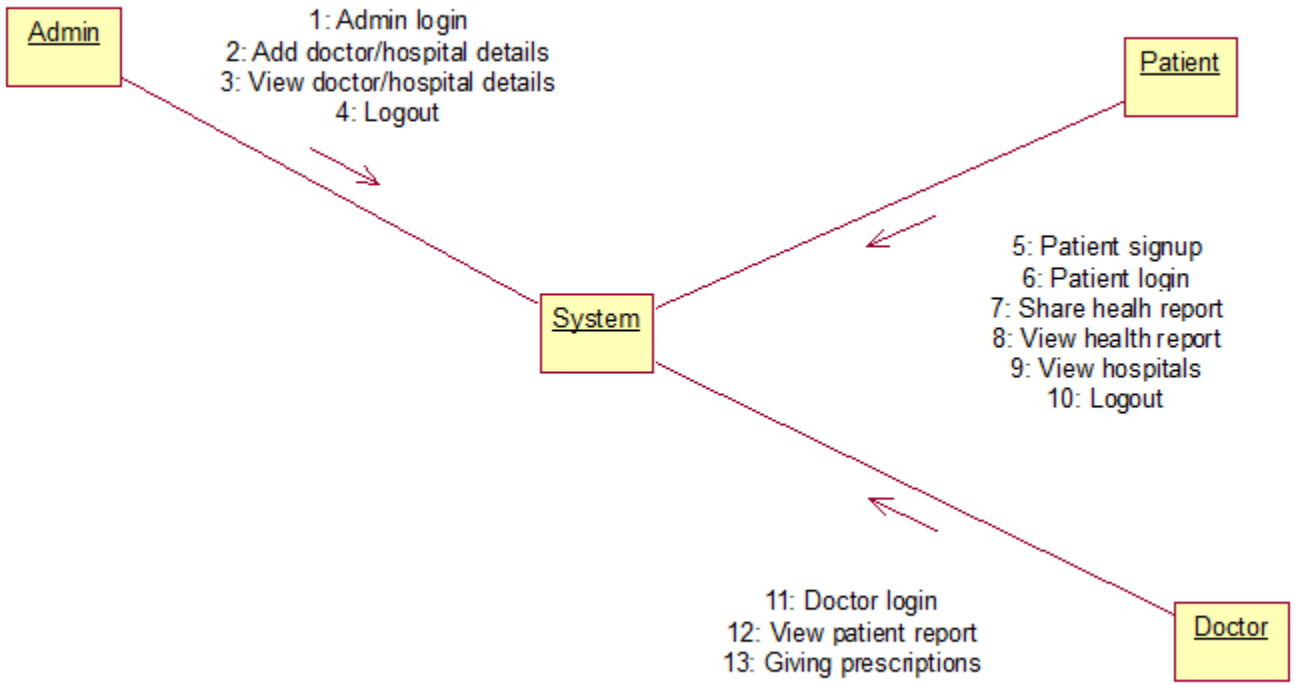
ACTIVITY DIAGRAM



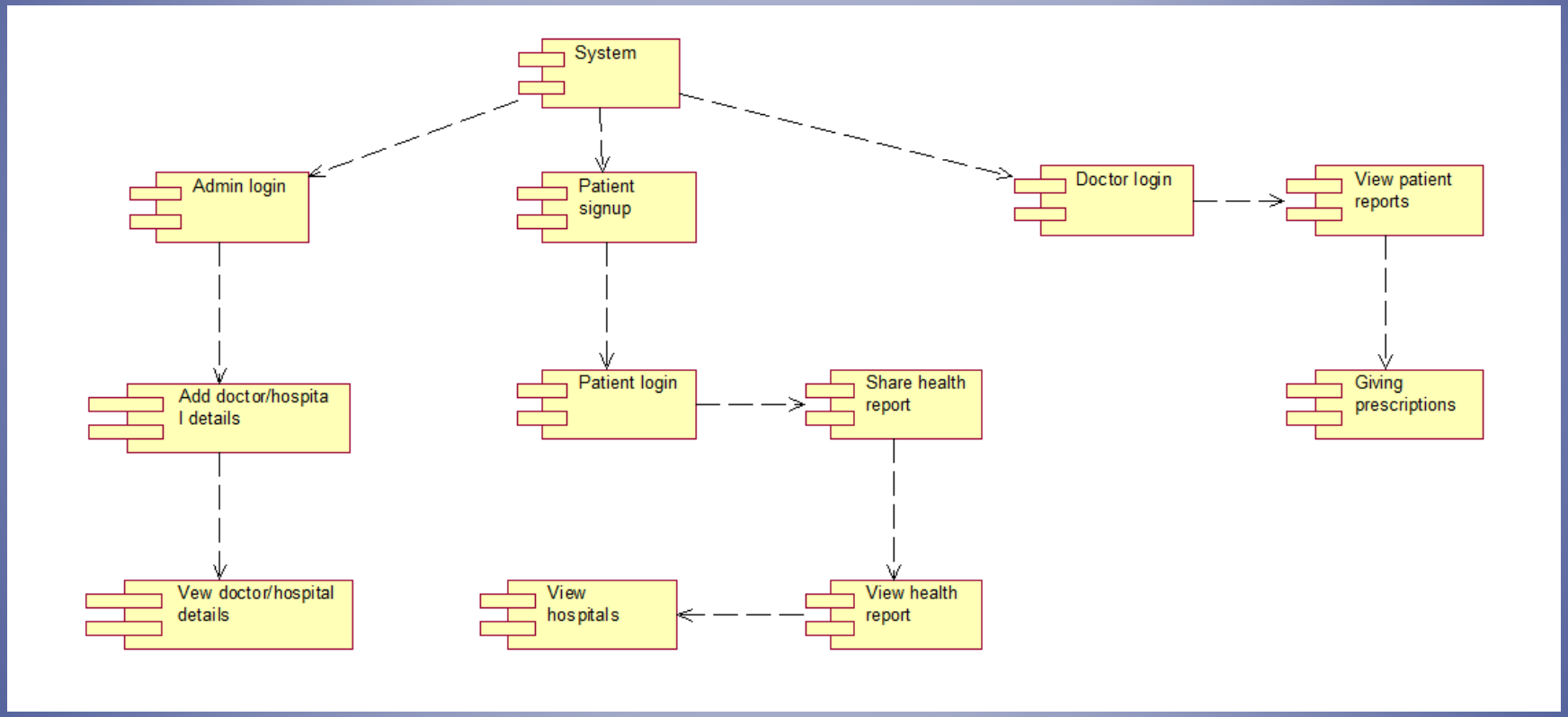
SEQUENCE DIAGRAM



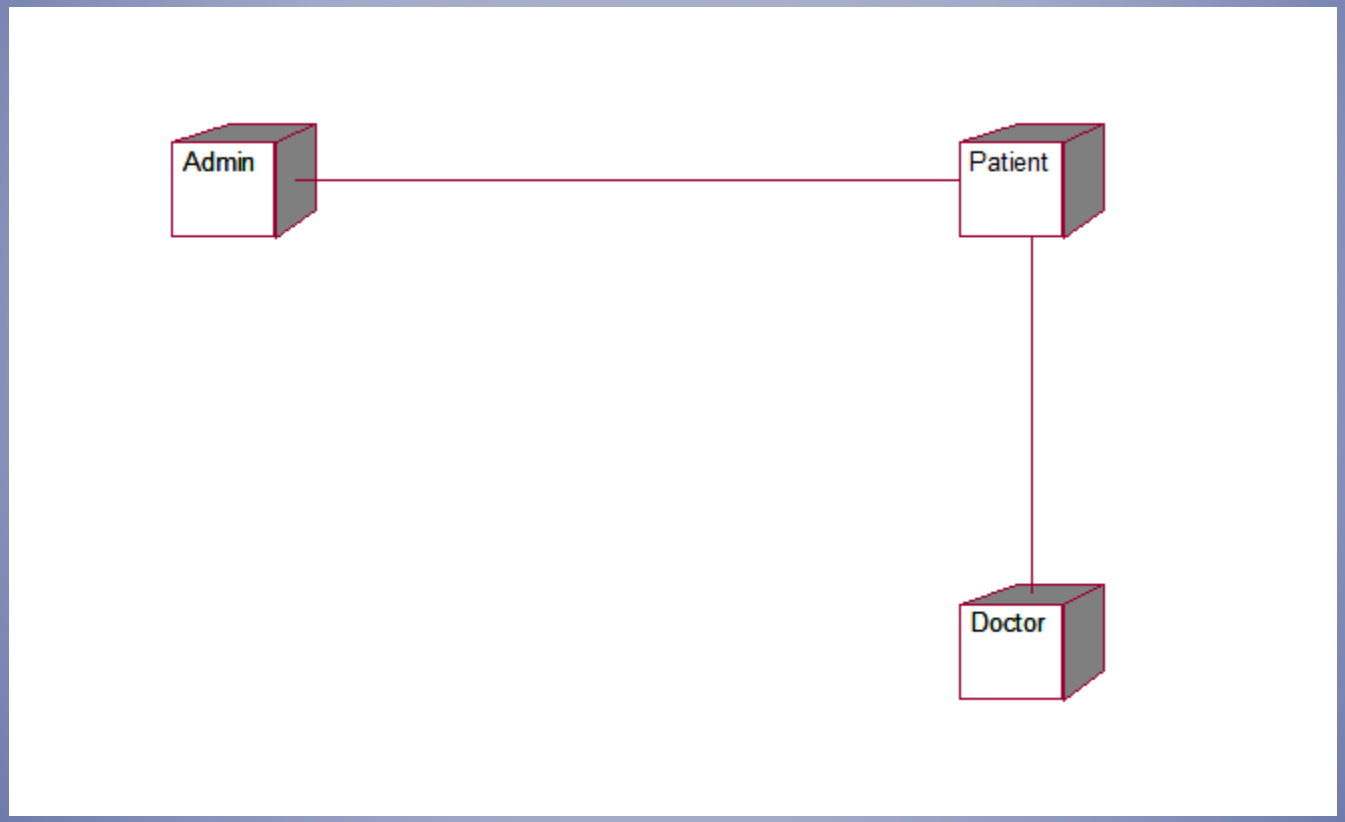
COLLABORATION DIAGRAM



COMPONENT DIAGRAM



DEPLOYMENT DIAGRAM



MODULES

- Admin Login:
- 1. Add Doctor/Hospital Details: The admin module allows authorized personnel to add and manage doctor and hospital details. This includes adding new doctors or hospitals to the system, updating their information, and ensuring their records are accurate.
- 2. View Doctor/Hospital Details: In this section, the admin can view the complete list of doctors and hospitals registered in the system. They can access their profiles, contact information, and any other relevant details.
- New Patient Signup:
- 3. Patient Signup: New patients can create accounts by providing their personal information. This typically involves entering their name, contact details, and creating a secure login. Once registered, they gain access to their patient dashboard.

- Patient Login:
- 4. Share Health/Medical Report: Registered patients can log in to their accounts and upload their health or medical reports. These reports may include diagnostic test results, medical history, or other health-related documents. After uploading, patients can share these reports with their healthcare providers or specialists.
- 5. View Health Report: Patients can access and review their own health reports, which are securely stored in their accounts. This allows them to monitor their medical history and share it with doctors or hospitals as needed.
- 6. View Hospitals: Patients can search and view a list of hospitals available in the system, access their contact details, and find information about the services they offer.
- Doctor Login:
- 7. View Patient Report: Doctors can log in to their accounts and access the health reports shared by patients. This feature enables them to review a patient's medical history and diagnostic records, aiding in diagnosis and treatment decisions.
- 8. Giving Prescription: Doctors have the ability to prescribe medications or treatments directly through the system. They can generate electronic prescriptions that are accessible to patients, making it more convenient and efficient for both doctors and patients.

CONCLUSION

- 1. The project introduces a transformative approach by integrating blockchain technology, addressing critical issues in healthcare data management and security.
- 2. Through the implementation of blockchain, the project establishes elevated security standards, ensuring the confidentiality and integrity of sensitive patient information.
- 3. Blockchain facilitates secure communication between doctors and patients, streamlining interactions and improving the overall efficiency of healthcare processes.
- 4. Medical records stored on the blockchain are rendered tamper-resistant, providing an immutable ledger that enhances trust and reliability in healthcare data.
- 5.With a foundation built on blockchain, the project sets the stage for a future-ready healthcare ecosystem that prioritizes transparency, security, and seamless collaboration among healthcare stakeholders.

REFERENCES

[1] King Yip, “BlockChain and alternative payment models,” White paper, 2016.

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[3] Li, H., et al., Blockchain-Based Data Preservation System for Medical Data J J. Med. Syst.2018. 42(8): p. 1- 13.

[4] Nguyen, D.C. , Pathirana, P.N. , Ding, M. , Seneviratne, A. , 2019. Blockchain for secureehrs sharing of mobile cloud based e-health systems. IEEE Access .

[5] Nitesh Gupta, AnandJha, and Purna Roy, “Adopting Blockchain Technology for ElectronicHealth Record Interoperability”, 2016.